Giustacchini 2017 CML — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=6) integrates all 35 plates: cross-sample neighbour fraction 0.82; one connected component, 0 islands removed. Read on the regenerated embedding across an n_pcs/θ sweep, not a scalar: θ=6 folds the blast-crisis patient triad into the body (vs deleting ~180 cells at θ=2).

Gate 2 — module geography → GO (non-branching)

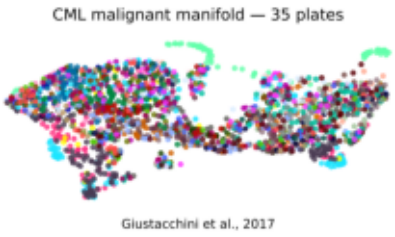
Non-branching case: a single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-vary (smoothed r=+0.70), both oppose Crest (Gast↔Crest r=−0.93, pSom↔Crest r=−0.80). Along pseudotime Gast falls (ρ −0.57), Crest rises to a terminal tip (ρ +0.60, Crest-high decile pt=0.90). Joint pSom&Crest enrichment 0.27 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — hub cross-repression → NA

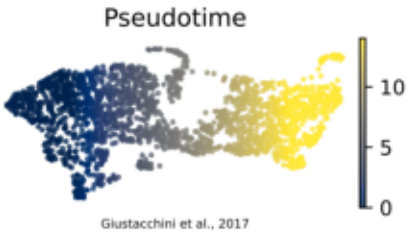
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks not produced, per spec).

STEP FIGURES

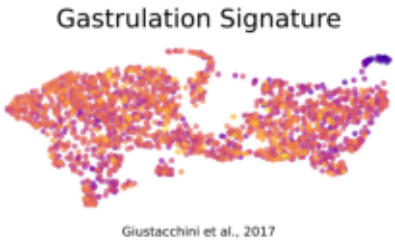
Manifold — samples



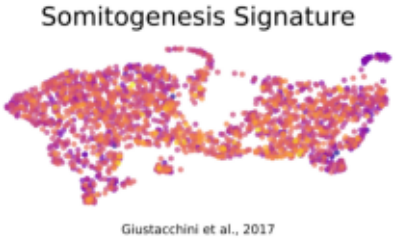
Pseudotime



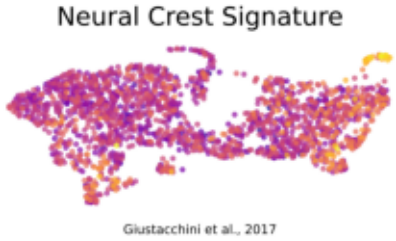
Gastrulation



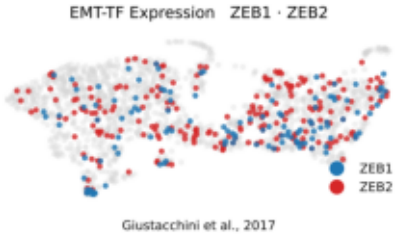
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
CML (Giustacchini et al., 2017), SmartSeq2 — 2,055 malignant BCR-ABL cells over 36 patient plates; single low-cycling body (~16% cycling), one continuous developmental axis.

Verdict.
Pattern reproduces in the non-branching Linear-Gast form: one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB1 & ZEB2 (only factors clearing detection), rising toward the Crest terminus.

Parameter choices

Sample threshold	n≥20 cells (35/36 plates, 2,040); ≥100 floor rejected (SmartSeq2 caps at 154); gap 15→22	**Overdispersed genes**	5,909 (scFates find_overdispersed)
Integration	Harmony on sample, θ=6 (stability sweep; folds blast-crisis patients into body)	**Embedding**	n_pcs=20 (scree elbow; n_pcs=40 fragments a patient speck)
Geometry	Non-branching scFates curve N=20 λ=0.3 μ=100; 2 tips/0 forks, SA 2.97; corr(x_lin,pt)=+0.99	**Root**	curve tip 0 = Gast/pSom-high, Crest-low origin
Module means (raw)	Gast +0.016 ≥ pSom +0.013 > Crest −0.016 (Linear-Gast); cell-cycle-free	**EMT-TFs shown**	ZEB1 (23%) & ZEB2 (42%); FOXC1/PRRX1/TWIST1/SNAI1 <2%, gated out